Developing an Artificial Intelligence (AI) model for the prediction of disease phenotypes using microbiome data

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The human gut microbiome plays a major role in maintaining host health through functions such as digestion, immune regulation, and metabolic control. Disruptions in microbial community composition, known as dysbiosis, have been associated with several diseases, including obesity, liver cirrhosis, and type 2 diabetes (T2D) etc. Previous studies utilized abundance tables as inputs for machine learning models to predict disease phenotypes. However, as microbial taxa exhibit phylogenetic and ecological relationships, graph-based deep learning methods are theoretically appealing because they can take these relationships into account, although their application to microbiome-based disease prediction remains limited. Therefore, this study evaluated and compared the performance of traditional machine learning, deep learning, and Graph Convolutional Network (GCN)-based models for microbiome disease prediction. Six publicly available gut microbiome datasets downloaded from the Metaml repository, representing liver cirrhosis, IBD, CRC, obesity, and two T2D cohorts, were analysed. Seven classification models were implemented, including Random Forest (RF), Support Vector Machine (SVM), Multilayer Perceptron (MLP), One-Dimensional Convolutional Neural Network (1D-CNN), and three GCN variants based on phylogenetic relationships, microbial interaction networks using the SPEC-EASI method, and their combination. Performance was assessed using 10-fold cross-validation and evaluated using accuracy, precision, recall, F1-score, AUROC, AUPRC, and F1-max. Comparative analysis demonstrated that the RF model consistently achieved the highest predictive performance across all six disease datasets, outperforming both deep learning and GCN-based approaches based on evaluation metrics. In contrast, the GCN models exhibited comparatively poor predictive performance and often failed to produce reliable classifications. These findings suggest that, despite the theoretical advantages of graph-based learning, current GCN implementations for microbiome classification remain limited by factors such as dataset size, graph quality, and network construction strategies. Overall, RF emerged as the most robust model for microbiome-based disease prediction and serves as a benchmark for future multimodal disease prediction systems integrating microbiome, genetic, environmental, and clinical data.

Keywords: Gut microbiome, disease prediction, machine learning, graph convolutional network, Random Forest

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